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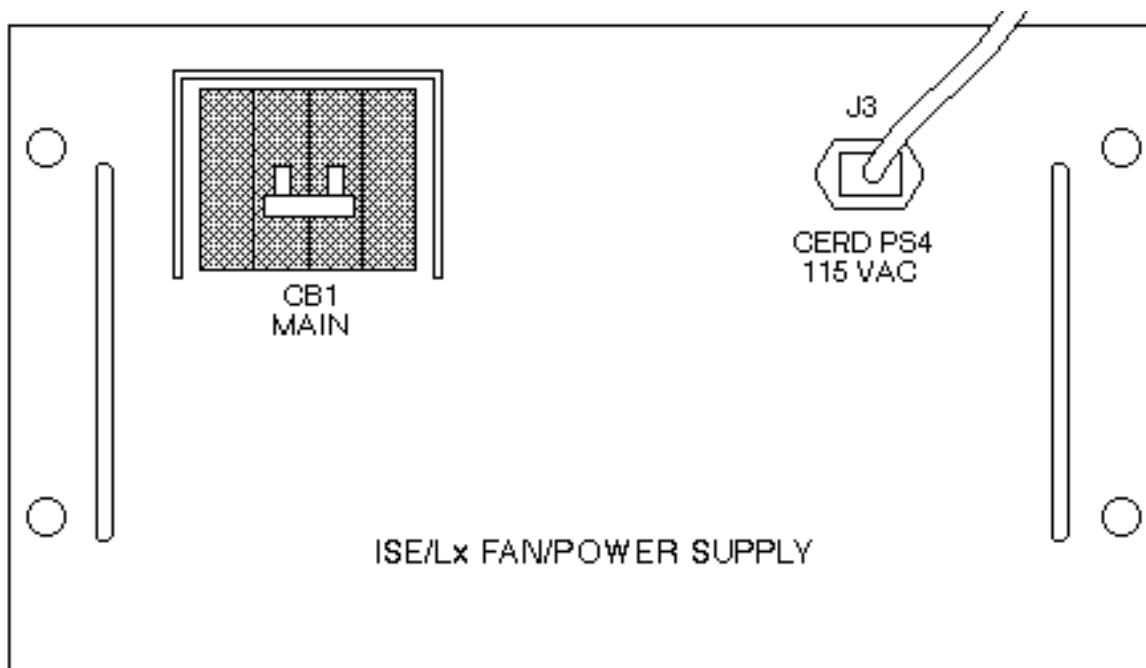
Description - Procedure for replacing the Integrated Pulse Generator (IPG) and (IPG II) Boards, located in the ISE/TPS Chassis.

Note

Except for the front panel layout, IPG and IPG II boards are 100% compatible.

1- REMOVAL OF FAILED BOARD

1. Remove front cover from System Cabinet.
2. Turn off the main circuit breaker, CB1 (see Illustration L4601A).



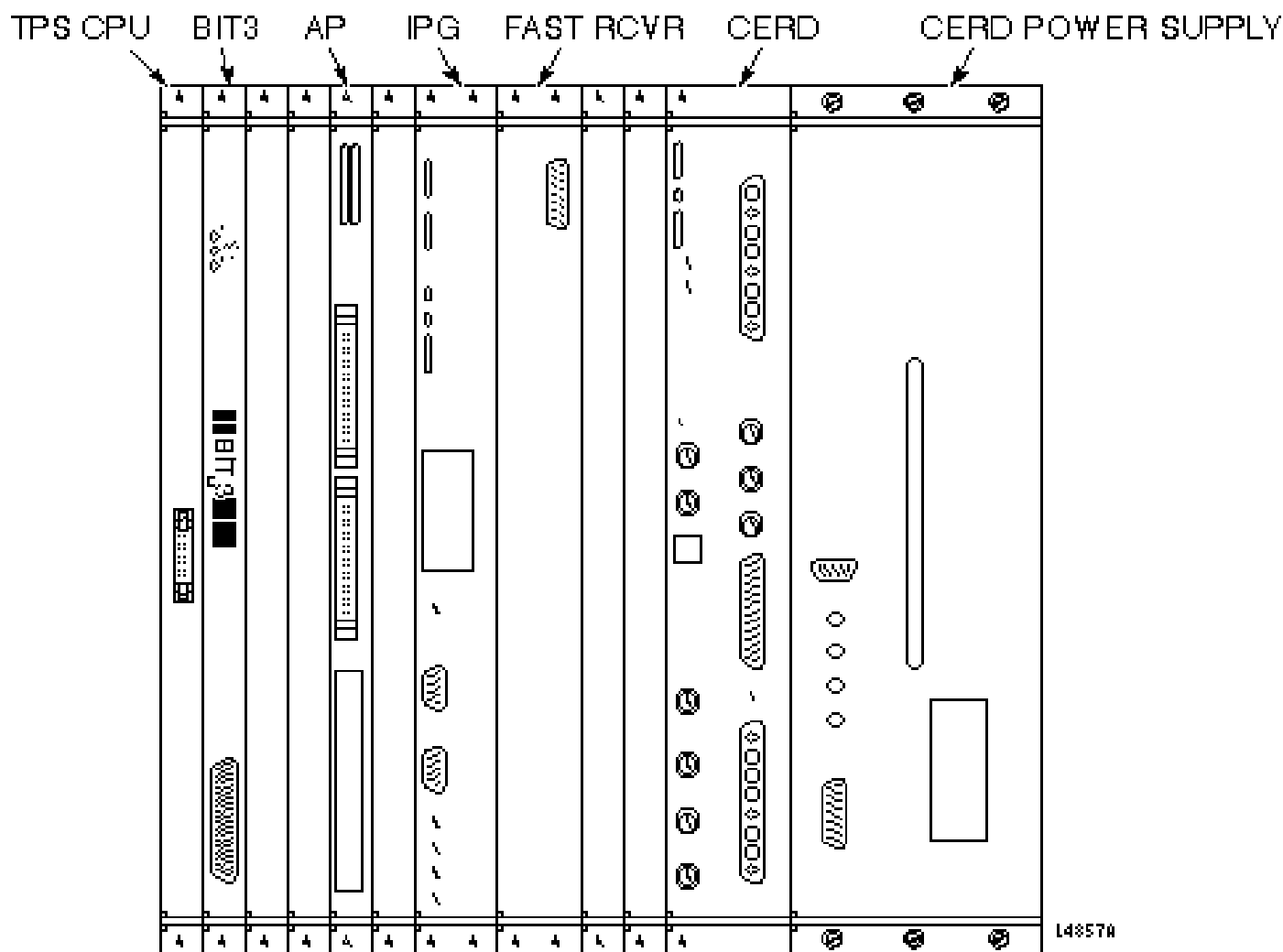
ISE/TPS FAN/POWER ASSEMBLY - FRONT VIEW
ILLUSTRATION L4601A



Equipment damage possibility.

Loosen the two front screws from the blank panel to the left (slot # 6) of the IPG Assembly and remove the blank panel. Components and RF fingers (EMC) on the IPG can be damaged if the blank panel is not removed.

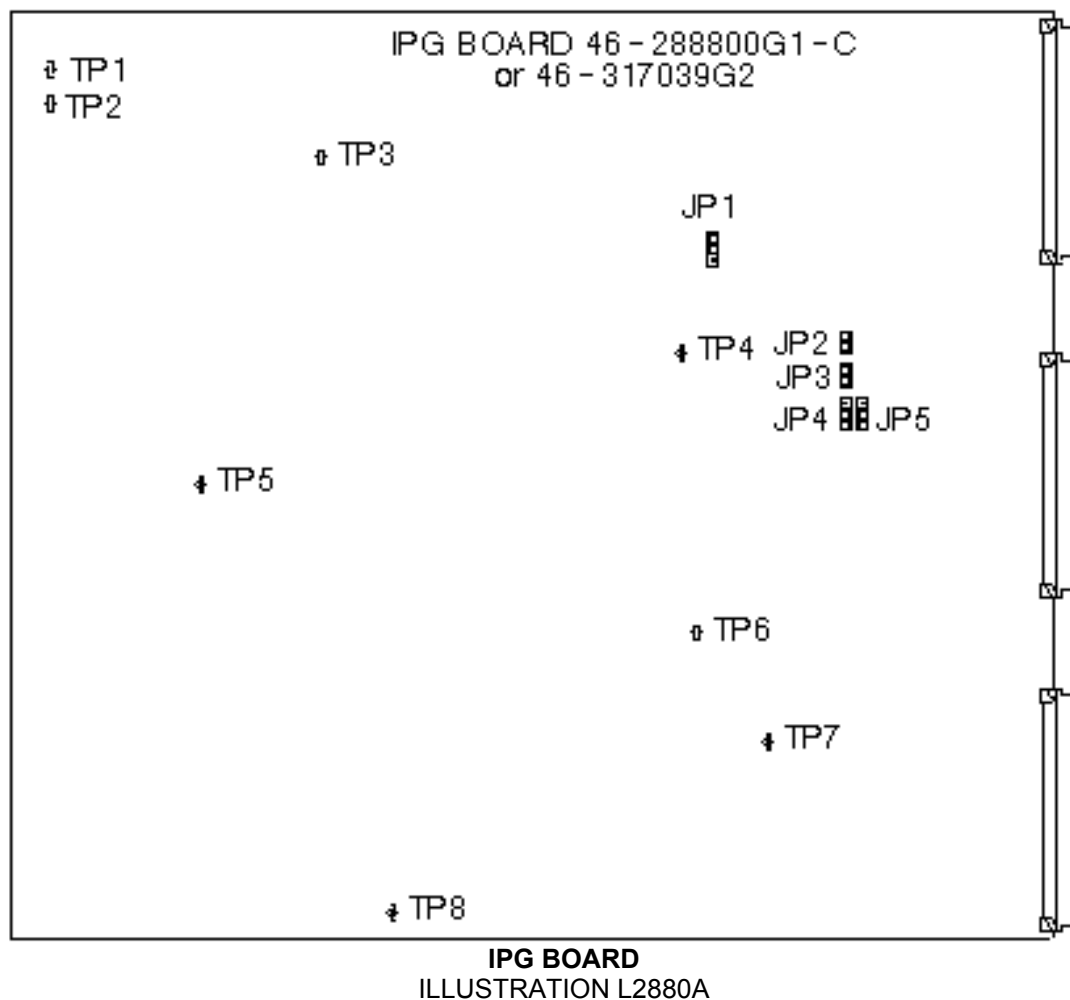
4. Loosen the two front panel screws. Remove the IPG Assembly from the ISE/TPS Chassis (see Illustration L4729A).



ISE/TPS CHASSIS - FRONT VIEW
ILLUSTRATION L4729A

2- BOARD SETUP

Illustration L2880A shows the IPG Board, along with its jumper locations and its test points.



See Table 1 for jumper information:

TABLE 1
JUMPER INFORMATION IPG

Jumper	Position	Description	Notes
JP1	1-2		
JP2	closed		
JP3	closed		
JP4	1-2		
JP5	1-2		

See Table 2 for test point information:

TABLE 2
TEST POINT INFORMATION IPG

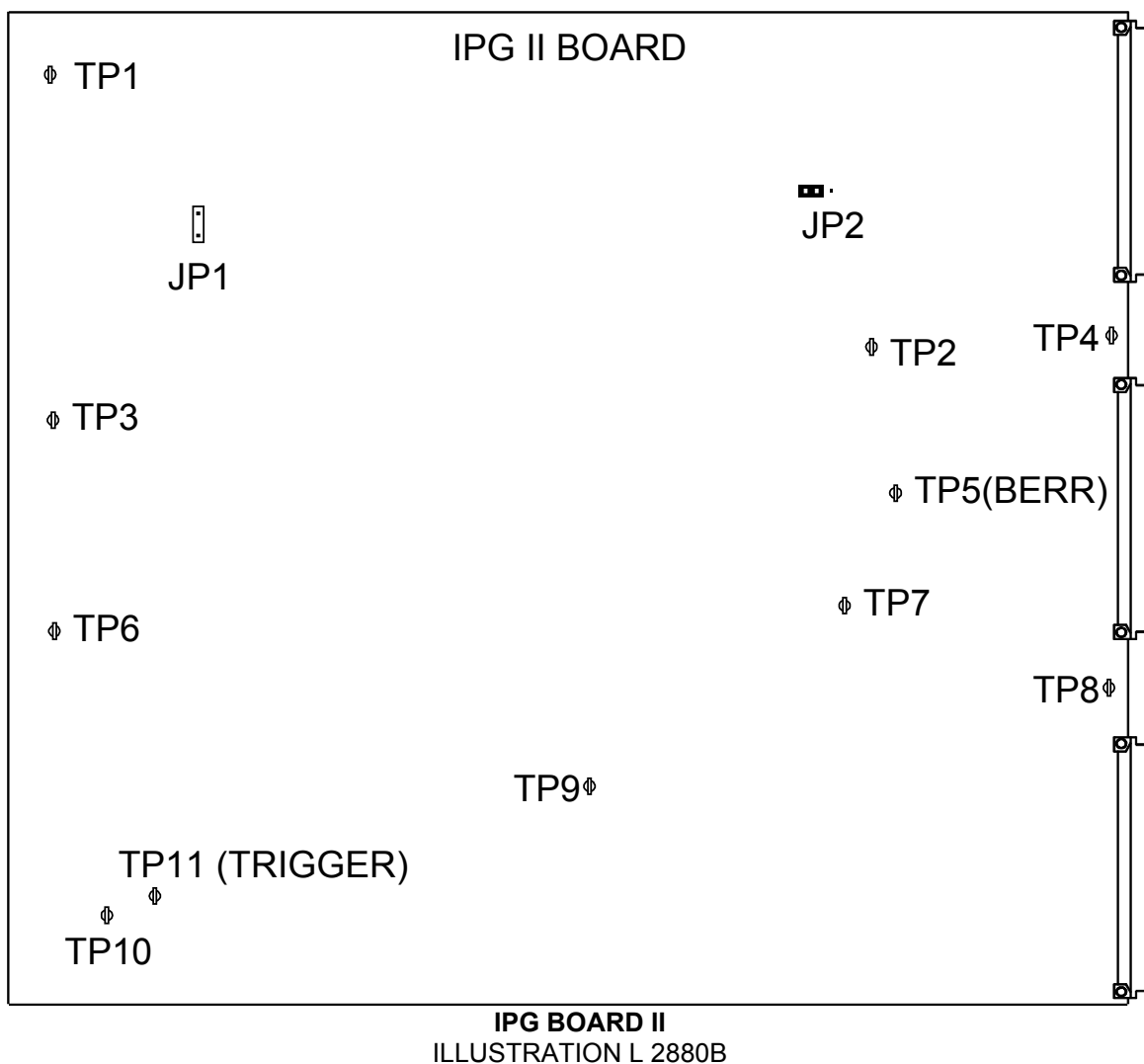
Test Point	Name	notes
TP1		GROUND
TP2	VCC	+5VDC
TP3		GROUND
TP4		B ERROR TEST POINT
TP5		GROUND
TP6	VEE	
TP7		GROUND
TP8		GROUND
J8	SCORE_TRIG*	FOREPLANE SMB
J9	RUFUNBLNK1*	FOREPLANE SMB
J10	RUFUNBLNK2*	FOREPLANE SMB
J11	UNBLANK1*	FOREPLANE SMB
J12	UNBLANK2*	FOREPLANE SMB

See Table 3 for LED information

TABLE 3
LED INFORMATION IPG

LED	Color	Name	Notes
DS1	Green	SSP PSG LED	
DS2	Green	X Gradient LED	
DS3	Green	Y Gradient LED	
DS4	Green	Z Gradient LED	
DS5	Green	Rho1	
DS6	Green	Rho2	
DS7	Green	Theta1	
DS8	Green	Theta2	
DS9	Green	WARP XFI	
DS10	Green	WARP LCA	
DS11	Green	SPU	
DS12	Green	SPI	
DS13	Green	Power Up Test	
DS14	Yellow	Disk Based Test	
DS15	Green	VME BusAccess	

Illustration L2880B shows the IPG II Board, along with its jumper locations and its test points.



See Table 4 for jumper information

TABLE 4
JUMPER INFORMATION IPG II

Jumper	Position	Description	Notes
JP1	open	normal operation	
JP2	1-2	flash write protect	

See Table 5 for test point information

TABLE 5
TEST POINT INFORMATION IPG II

Test Point	Name	notes
TP1		GROUND
TP2		GROUND
TP3		GROUND
TP4		GROUND
TP5		BERR
TP6		GROUND
TP7		GROUND
TP8		GROUND
TP9		GROUND
TP10		GROUND
TP11		XF TRIGGER
J6	SPI Serial Debug Port	
J7	68020 Serial Debug Port	
J8	SCOPE_TRIG*	FOREPLANE SMB
J9	RF UNBLNK1*	FOREPLANE SMB
J10	RF UNBLNK2*	FOREPLANE SMB
J11	UNBLNK1*	FOREPLANE SMB
J12	UNBLNK2*	FOREPLANE SMB

See Table 6 for LED information

TABLE 6
LED INFORMATION IPG II

LED	Color	Name	Notes
DS5	Green	Sequencers Operational	
DS6	Green	VME Access	
DS7	Green	SPI CPU	
DS8	Green	SPU CPU	
DS9	Green	WARP XF1 Interlock signal	Indicates WARP is ready to read gradient data
DS10	Green	68020 Power Up Test	

3- INSTALLATION OF REPLACEMENT BOARD

1. Install IPG Board into ISE/TPS Chassis and tighten the two front panel screws (see Illustration L4729A in Section 1).
2. Install the blank panel into the ISE/TPS Chassis and tighten the two front panel screws. (see Illustration L4729A in Section 1).
3. Turn on the main circuit breaker, CB1 (see Illustration L4601A in Section 1).
4. Install System Cabinet front cover.
5. **[Reset TPS].**

4- FUNCTIONAL CHECKS REQUIRED

1. Perform one head or body scan.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 1, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.
1	April 22, 1999	R. Kaufman	Corrected IPG-II jumper information